

T-SQL → BigQuery SQL — Dialect Cheat Sheet

BigQuery speaks **GoogleSQL** (standard SQL). Most of your T-SQL transfers directly; this card is the differences you'll actually hit, first used on Day 1 afternoon and hardest-needed in the Day-2 Iceberg lab. Every row below matches the BigQuery docs (links at the bottom).

The everyday rows

T-SQL	BigQuery GoogleSQL	Notes
[Order ID]	`Order ID`	Backticks, not square brackets. Quote the whole three-part name as one: `project.dataset.table`
SELECT TOP 100 *	SELECT * ... LIMIT 100	LIMIT, not TOP — and it goes at the end of the query
ISNULL(col, 0)	IFNULL(col, 0) or COALESCE(col, 0)	Prefer COALESCE (standard, takes multiple fallbacks)
CAST(x AS INT) (errors on bad data)	SAFE_CAST(x AS INT64)	SAFE_CAST returns NULL instead of failing the query — put it in Silver-layer models that absorb vendor-file dirt
GETDATE()	CURRENT_TIMESTAMP()	Returns TIMESTAMP (UTC). Also CURRENT_DATE(), CURRENT_DATETIME()
DATEADD(day, -7, d)	DATE_SUB(d, INTERVAL 7 DAY)	Add with DATE_ADD, subtract with DATE_SUB; also TIMESTAMP_ADD / DATETIME_ADD / TIMESTAMP_SUB
DATEDIFF(day, a, b)	DATE_DIFF(b, a, DAY)	Argument order flips: later date first. Also TIMESTAMP_DIFF
LEN(s)	LENGTH(s)	
SUBSTRING(s, 1, 3)	SUBSTR(s, 1, 3)	SUBSTRING also works as an alias; positions are 1-based in both
CHARINDEX('x', s)	STRPOS(s, 'x')	Haystack first, needle

T-SQL	BigQuery GoogleSQL	Notes
		second — the argument order flips
$s1 + s2$	<code>CONCAT(s1, s2)</code> or $s1 s2$	CONCAT treats NULL as empty string (no <code>CONCAT_NULL_YIELDS_NULL</code> setting to remember)
$5 / 2 \rightarrow 2$	$5 / 2 \rightarrow 2.5$	Division always returns a floating-point type. Integer division is <code>DIV(5, 2) → 2</code>
divide by zero → error	also an error	Use <code>SAFE_DIVIDE(x, y)</code> to get NULL instead
<code>int / bigint</code>	<code>INT64</code>	One integer type. <code>datetime</code> ≈ <code>TIMESTAMP (UTC)</code> or <code>DATETIME</code> (civil time, no zone) — pick deliberately per column
Stored procedures, batches	Scripting: <code>DECLARE x INT64; SET x = ...; plus IF ... END IF;, WHILE ... END WHILE;</code>	Multi-statement scripts run straight in the query editor — your "anonymous sproc" instinct works
<code>MERGE</code>	<code>MERGE</code>	Exists, same purpose: upsert by match condition
CTEs (<code>WITH</code>), window functions (<code>ROW_NUMBER()</code> OVER (<code>PARTITION BY ...</code>))	identical	Work exactly as you expect
<code>USE mydb</code>	—	No USE. Always fully qualify: <code>`project.dataset.table`</code>

Things that will bite you

1. **Case sensitivity.** String comparisons are case-sensitive (`'smith' ≠ 'Smith'`), and dataset and table names are case-sensitive by default. If your SQL Server collation is the usual case-insensitive sort, this is the difference most likely to produce "the same" rows not joining.

2. **Project IDs contain dashes — backtick the whole name.** Lab projects have IDs like `qwiklabs-gcp-01-ab12cd34ef56`. ``qwiklabs-gcp-01-ab12cd34ef56.class.orders`` works; unquoted or partially quoted versions fail.
3. **Integer math doesn't truncate, and zero divides loudly.** `/` returns a float; divide-by-zero errors out the whole query. Reach for `DIV()` and `SAFE_DIVIDE()`.
4. **Nested/repeated columns need UNNEST.** BigQuery tables can hold `ARRAY<STRUCT<...>>` columns (an order with its line items inside the row). You can't treat them like a child table until you flatten: `FROM orders, UNNEST(line_items) AS li`. You'll meet this in the derived-tables activity (DIN §29).
5. **TIMESTAMP ≠ DATETIME.** Comparing or joining the two types errors instead of implicitly converting. Decide per column: `TIMESTAMP` for event instants, `DATETIME` for "wall-clock" values like a business date.
6. **External-table DDL is metadata-only and returns instantly.** If a `CREATE EXTERNAL TABLE` seems to hang, the statement is wrong (usually the URI) — it's not "reading the files."

Pleasant surprises

- `SELECT * EXCEPT (cust_email, cust_phone)` — project everything *but* the PII, no column list.
- `SELECT * REPLACE (SAFE_CAST(minutes AS INT64) AS minutes)` — re-type a column inline.
- `QUALIFY` — filter on a window function without a subquery: `... QUALIFY ROW_NUMBER() OVER (PARTITION BY case_id ORDER BY updated_at DESC) = 1` — your "latest record per key" pattern in one line.

Dig deeper

- [Query syntax](#) (identifiers, `LIMIT`, CTEs, `UNNEST`, `QUALIFY`)
- [Conditional expressions](#) (`IFNULL`, `COALESCE`)
- [Conversion functions](#) (`SAFE_CAST`)
- [Date functions](#) · [Timestamp functions](#) · [String functions](#) · [Operators](#) (`DIV`, `SAFE_DIVIDE`)
- [Scripting](#) (`DECLARE`, `SET`, `IF`, `WHILE`)
- [DML](#) (`MERGE`)

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